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25 June 1984
SUPERSEDING
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4 October 1979

MILITARY SPECIFICATION

SHAPER, METAL CUTTING, HORIZONTAL RAM, HYDRAULIC PLAIN AND UNIVERSAL TABLE, SIZES 24 and 28

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers one type of metal cutting shaper having a horizontal ram, plain or universal table and hydraulic powered ram with equipment and accessories specified herein.

1.2 Classification. This specification covers shaper machines of the following sizes. The size required shall be as specified (see 6.2.1).

Size 24 - 24 inch cutting stroke
Size 28 - 28 inch cutting stroke

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-TES, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3418

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2. APPLICABLE. DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/IEEE 268 - Metric Practice.

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues.

ANSI/NFPA 79 - Metalworking Machine Tools and Plastics Machinery, Electrical Standard for

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM-D-3951 - Commercial Packaging, Standard Practice for.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First Article. When specified, (see 6.2.1) a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Design. The shaper shall be new and one of the manufacturer's current models capable of removing metal in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be us-cd in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in U.S. units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in ANSI/IEEE 268.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2,1). Regardless of the

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measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1 .1.1 Dual calibrations; When specified (see 6.2.1), meassuring and indicating devices shall be graduated in both the US and the SI System of Measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials provied such materials will not jeopardize the machine's intended use and performance The reclaimed materials shall have been reprocessed, remainufactured or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The shaper and all its applicable components that directly consume -energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments availble within the Industry.

3.2.4 Controls. All operating controls shall be located convenient to the operator at his norms-l work station.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the shaper that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety, devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.6 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease-fittings and filler caps shall be accessible.

3.2.7 Interchangeability. To provide for replacement of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The shaper shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable

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of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the shaper shall be painted in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials that indicate stock removal or tool movement shall be graduated in increments not larger than .001 inch. Dial diameters shall be such that graduations may be read from the normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.4 Components.

3.4.1 Base and column. The base and column shall have strength and size to assure maximum rigidity under capacity load. The structure shall be stress relieved and normalized to minimize distortion and deflection. The column shall have horizontal ways on its upper surface for supporting the ram and shall have vertical ways on its frontside for supporting the crossrails and table. The length and width of the ram ways or bearings in the column shall conform to table I. The base of the machine shall have a way or slide for the table outboard support. The column shall have covered

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access openings of ample quantity and size to allow servicing of all components contained within the assembly. Means shall be provided for leveling the machine and for securely bolting it to a foundation.

3.4.2 Ram. The ram shall be a one-piece casting having ways which are machined and ground or handscraped to fit the ways in the column. The ways shall have means for adjustment to compensate for wear. The ways shall-be automatically lubricated by a filtered pressurized lubrication system. The ram shall be hydraulically driven by a ram drive mechanism conforming to 3.4.3, and shall have a tool head conforming to 3.4.8. The length of the ram stroke, number of ram cutting and return speed selections, and the range of cutting and return speeds shall conform to table I.

3.4.3 Ram-drive system. The ram shall be hydraulically actuated by a system consisting essentially-of a hydraulic fluid reservoir, pump, pressure control and relief valves, necessary piping and tubing, piston and cylinder, and a reversing valve. The hydraulic reservoir shall either be located in the base of the machine or be a separate tank. The hydraulic pump shall have sufficient pressure and flow capacity to achieve the cutting speeds specified in table I. Relief valves shall be provided to protect the system during overloads. A reversing valve operated by adjustable position trips shall be provided for reversing the ram at the end of each stroke. Means shall be provided for adjusting the cutting speed, return speed and the position stroke with the ram either stopped or in motion without damaging the machine or its components. Means shall be provided to manually short-stroke the ram at any location within the established stroke without disturbing the position of the trips.

3.4.4 Crossrail. The crossrail shall have a horizontal way which supports the table and allows horizontal movement of the table. The crossrail shall also have a vertical way slide which is fitted to the vertical way of the column, allowing the crossrail and table to be elevated and lowered. The ways and slides shall be machined and ground or handscraped, and shall have means of adjustment to compensate for wear. The crossrail length shall be such that the table can travel horizontally the distances specified in table I. The crossrail shall have clamps that will securely hold the crossrail to the column ways at any adjusted height position.

3.4.4.1 Crossrail elevating system. The machine shall have a manually operated elevating screw which will allow the crossrail and the attached table to be manually elevated through the table vertical travel distance specified in table I. The machine shall also have a powered elevating system that will raise and lower the crossrail and table at a rate not less than 24 inches per minute. The powered elevating system may be either hydraulically or mechanically operated.

3.4.5 Tables. Unless otherwise specified (see 6.2.1), the machine shall be furnished with a plain table. When specified, a universal table shall be furnished.

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3.4.5.1 Plain table. When a plain table is furnished, the table shall be fixed, non-revolving and non-tilting. The table shall have an apron which has a machined and ground or handscraped way slide fitted to the horizontal ways of the crossrail, allowing the table to move horizontally along the crossrail way. The table shall be securely attached to, but shall be removable from, the apron to allow large workplaces to be attached directly to the apron. The work supporting surface of the apron shall have not less than two T-slots conforming to ANSI B5.1. The table top surface and both side surfaces shall be machined with the side surfaces square to the top surface within an accuracy of 0.001 inch per 12 inches. The top surface shall have not less than three T-slots conforming to ANSI 65.1. The table top surface shall have not less than three T-slots, and the side surfaces shall have not less than two T-slots, all conforming to ANSI B5.1. The table shall have a removable outboard support attached to the free end of the table which slides on a guide or way attached to the base. The height of the outboard bearing support shall be adjustable for use at any table height within the range specified herein.

3.4.5.2 Universal table. When a universal table is specified, the table shall have a tilting top surface and shall rotate about-a horizontal axis perpendicular to the-plane of the column ways. The table apron shall have a way slide which is machined and ground or handscraped to fit the horizontal way of the crossrail. The apron shall have a trunnion which supports the table and allows the table to be rotated not less than 360° about the trunnion axis. The top surface of the table shall tilt about a horizontal axis parallel to the crossrail axis through an angle not less than 15° above and below the horizontal plane. Both the rotating table and the tilting top surface shall have a scale, graduated in increments not larger than 1 degree, to indicate the angles of rotation and tilt, and shall have cranks for rotating the table and tilting the top surface. Means for securely clamping the rotating table and tilt top in any desired angular position within the specified ranges shall be provided. The top surface and the right side surface of table shall be machined square to each other within 0.001 inch per 12 inches. Both the tilting top surface and the machined side surface shall have not less than three T-slots conforming to ANSI B5.1. The table shall have a removable outboard support which is attached to the outboard end of the trunnion and which slides on a guide or way on the machine base. The height of the outboard support shall be adjustable for use at any table height within the range specified herein.

3.4.6 Table horizontal feed mechanism. The machine shall have provisions for feeding the table in both horizontal directions, both manually and by power. The manual feed system shall consist of handwheels or cranks and feed dials located at both ends of the crossrail to enable the operator to feed and position the table from either location. The feed dials shall be graduated in increments not larger than 0.001 inch and shall indicate the amount of table movement. The power feed system shall be hydraulically actuated, shall operate on the end of the return stroke of the ram, and shall provide the table feed rates specified in Table I.

3.4.7 Table rapid traverse mechanism. The machine shall have a rapid traverse mechanism for the horizontal movement of the table. The mechanism shall traverse the table horizontally on the crossrail in both directions at

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a rate not less than 24 inches per minute. The rapid traverse mechanism shall incorporate means to prevent damage if the table is inadvertently traversed or fed to the extreme ends of the table travel.

3.4.8 Tool head. The tool head shall be of the swivel type having not less than 90 degrees of angular rotation to each side of vertical position. "The tool head shall have a scale graduated in increments not larger than one degree, to indicate the number of degrees the head is swiveled off the vertical position. Means shall be provided to securely clamp the tool-head in any angular position within its swivel range. The tool slide shall have a manually operated handwheel or crank, and a feed screw mechanism which will downfeed the tool the distance specified in table I. When specified (see 6..2.1), the tool head shall have a powered downfeed system in addition to the manual feed system specified above. The powered downfeed mechanism shall provide feeds ranging from 0.005 inch or less, to 0.018 inch or more, per stroke. The manual feed mechanism and the powered feed mechanism, when furnished, shall be capable of feeding the tool slide vertically or at-any swivel angle to which the slide may be adjusted. The tool slide shall have a hinged swivel type clapper box supporting a tool post with a slot of the size specified in table I. The clapper box shall swivel not less than 15 degrees to each side of vertical position. Unless otherwise specified (see 6.2.1), an automatic tool lifter shall be furnished.

3.4.9 Vise. Unless otherwise specified, the vise jaws shall be operated by a single screw, and shall swivel not less than 90 degrees' either side of the table axis. Means for secure clamping at any swivel angle shall be provided. The vise or vise base shall be graduated in increments not larger than one degree to indicate the swivel angle. When specified (see 6.2.1), the moveable jaw of the vise shall be operated by two screws, and shall swivel to facilitate clamping of tapered workplaces. When specified (see 6.2.1), the vise jaws shall have replaceable steel inserts. The vise base shall have a locating key or rail that fits the center T-slots in the table work surfaces to align the axis of the vise parallel to the work table axis. The vise dimensions shall conform to table 1.

3.4.10 Coolant system. The coolant system shall include a sump or reservoir, a power driven pump and all necessary piping. The sump or reservoir shall have sufficient capacity to permit full flow of coolant. The system shall have means for draining and cleaning and shall include a strainer which is easily removable for cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant.

3.4.11 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

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3.4.12 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA-79. Each machine shall draw all of its electrical- power from a single 230/460 volt, 3 phase, 60 Hz circuit. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 230 volts, An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for. connecting the size grounding conductor specified in ANSI/NFPA-79 for the disconnect fuse rating.

3.4.13 Motors. Motors shall have ball or roller bearings, and shall conform to the requirements of table 1. Unless otherwise specified (see 6.2,1), each motor enclosure Shall meet the requirements of a dripproof enclosure.

3.5 Size and characteristics. The size and characteristics of the shaper shall meet the requirements of table I.

3.6 Performance. Each machine shall be capable of performance as specified in table I and table 11. Each machine shall be capable of rough and finish machining the workpiece at the cutting speed, depth of cut, and crossfeed rate specified in table II.

3.7 Alignment tolerances. Each machine shall conform to the alignment tolerances specified in table III.

3.8 Standard equipment. One set of hand tools, normally furnished by the manufacturer, shall be furnished.

3.9 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.10 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following Information: .

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

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TABLE I. Sizes and capacities.

CHARACTERISTICS	REQUIREMENTS	
	Size 24	Size 28
Ram:		
Length of stroke:		
Nominal rating, inches	24	28
Actual rating, inches	1 to 26	1 to 30
Ram bearing in column:		
Length, in inches	40	45
Width, inches	12-3/4	14-1/4
Cutting speeds:		
No. of speed selections	inf. variable	inf. variable
Range, feet per minute	10 - 140	10 - 100
Return speeds:		
No. of speed selections	inf. variable	inf. variable
Range, feet per minute	20 - 150	20 - 150
Table:		
Cross-feed		
No. of selections	inf. variable	inf. variable
Range, inches per minute	0.002 to 0.110	0.002 to 0.110
Table travel:		
Horizontal, inches	26	30-1/2
Vertical, inches	10-3/8	13-1/8
Ram to table surface, max, distance:		
Plain table, inches	14-3/4	17-1/2
Universal table		
Tilt surface, inches	14-1/8	16-5/8
Plain surface, inches	16-1/8	18-5/8
Tool Head:		
Tool head travel, without power feed, inches	8	8-1/4
With power feed, inches:	7	7
Tool post slot size, inches:	7/8 x 1-7/8	1 x 2
Vise:		
Vise jaw size, inches	2-1/2 x 15	3 x 17
Vise opening maximum, inches	14	15
Vise swivel either direction of axis, degrees	90	90
Motor HP	7-1/2	10

NOTE: Requirements are "not less than" those shown. Where a range is shown, the performance shall encompass the range from the stated "minimum or less" to the stated "maximum or greater."

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TABLE II. Performance characteristics.

Machine Size	Cutting speed Ft. Per Min., Not Less Than	Depth of Cut Inches, Not Less Than	Cross-Feed Rate, Inches Per Stroke Not Less Than	Allowable Thickness, Variation, Inches
Rough Cut:				
24	50	0.350	0.035	0.011
28	50	0.350	0.035	0.012
Finish Cut:				
All Sizes	90	0.015	0.010	0.002

TABLE III. Alignment Tolerances

ALIGNMENT CHARACTERISTICS	TOLERANCES
Ram travel parallel to table top surface (free-end of ram closer to table in ram extended position)	0.001 inch in 12 inches
Ram travel parallel to table side surfaces	0.001 inch in 12 inches
Ram travel parallel to table top center T-slots	0.001 inch in 12 inches
Ram travel parallel to top of vise jaw way	0.001 inch in 12 inches
Rise and fall of table during horizontal travel	0.0005 inch in 12 inches
Vertical travel of table parallel to travel sides	0.001 inch in 12 inches
Tool slide travel square with table top (with slided adjusted to 0° position)	0.001 inch in 12 inches

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3.10.1 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include-the model of the basic standard machine and a suffix Identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturers model designation
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract.

3.12 Workmanship . Workmanship of the shaper and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or Purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies-and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).-
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the item to pass the first article tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1) quality conformance inspection shall consist of at least the examination in 4.5 and the tests in 4.6.2. Failure of the item to pass the examination or tests shall be cause for rejection.

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4.5. Examination. The machine and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.5 and 3.6 through 3.12.

4.6 Tests.

4.6.1 Operational test. The shaper shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.6.2 Alignment accuracy tests. Each shaper shall be tested to determine that it complies with the alignment accuracy requirements of table III (see 3.7).

4.6.3 Performance tests.

4.6.3.1 Performance test No. 1. The machine shall be subjected to the following cutting force test. A calibrated pressure gauge shall be installed in the-hydraulic system to indicate hydraulic pressure at the inlet of the hydraulic cylinder that operates the ram head. A workpiece of suitable size shall be installed on the table, a suitable cutting tool shall be installed in the tool holder, and the ram speed shall be adjusted to the maximum cutting speed shown in table II. A cut shall be started, and the cutting depth and feed shall be increased until the gauge reading at inlet to hydraulic cylinder equals 90% of rated pressure without the ram stalling.

Rate Ram Cutting Force = (.95) x (maximum pressure gage reading) x (total working area of hydraulic piston as calculated from certified manufacturing drawings.)

4.6.3.2 Performance test No. 2. The ram shall be stalled in the dashpot of the cylinder in the cutting direction at low speed. The gauge reading shall be equal to rated pressure necessary to obtain maximum cutting force. During this test there shall be no evidence of hydraulic fluid leakage, overheating of motors, or permanent distortion of any machine component or structural members.

4.7 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified, packaging, packing, and marking shall be in accordance with ASTM-D 3951. When specified (see 6.2.1), level A packaging, level A or level B packing and special marking shall be accomplished in accordance with MIL-M-18058.

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6, NOTES

6.1 Intended use. The shapers covered by this specification are intended for rough and finish machining of metal workplaces.

6.2 Ordering data.

6.2.1 Procurement requirements. Purchaser specify their-requirements in procurement documents, including whether each choice is required or not required, by entering an appropriate statement identified to each of the following:

a. Title, number, and date of this specification.
b. Size machine required (see 1.2).
c. When a first article is required for inspection and approval (See 3.1, 4.3, and 6.3).
d. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).
If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI) state required system (see 3.2.1.1).

f. Dual calibrations (US and SI); if required (see 3.2.1.1.1).
g. Specify additional safety and health requirements (see 3.2.5).
h. Painting, if different (see 3.3.5).
i. When a universal table shall be furnished (see 3.4.5).
j. Powered downfeed system to tool head, if required (see 3.4.8).
k. When automatic tool lifter is not required (see 3.4.8).
1. Double screw vise, if required (see 3.4.9).
m. Vise jaw inserts, if required (see 3.4.9).
n. Electrical system, if different (see 3.4.12).
o. Type motor enclosure, if different (see 3.4.13).
p. Optional equipment, if required; fully described required optional equipment (see 3.9)$_0$
r. Lubrication chart or plate, if different (see 3.10).
s. Nameplate if different (see 3.10.1).
t. First article inspection, if different (see 4.3).
u. Quality conformance inspection, if different (see 4.4).
v. Preservation, packing and marking, if different (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operators manuals, parts lists, wiring diagrams, examination, alignment and tolerance test reports and other instructions for operation and maintenance (as identified on a DD Form 1664, Data Item Description) should be specified on a DD Form 1423, Contract Data Requirements List incorporated into the' contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test and approval of the first article.

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6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. The machine-shall be equipped with all points-of-operation guarding normally furnished as standard on the manufacture's commercial machines supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, as in most cases, the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements should be specified in detail under 6,2.1,

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing Activity:
DLA-IP

Project Number:
3418-0027

Review Activities:
Air Force - 84
DLA -GS

User Activities:
Army - SM
Navy - SH

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-23375D	Shaper, Metal Cutting, Horizontal Ram, Hydraulic Plain and Universal Table, Sizes 24 and 28

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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